

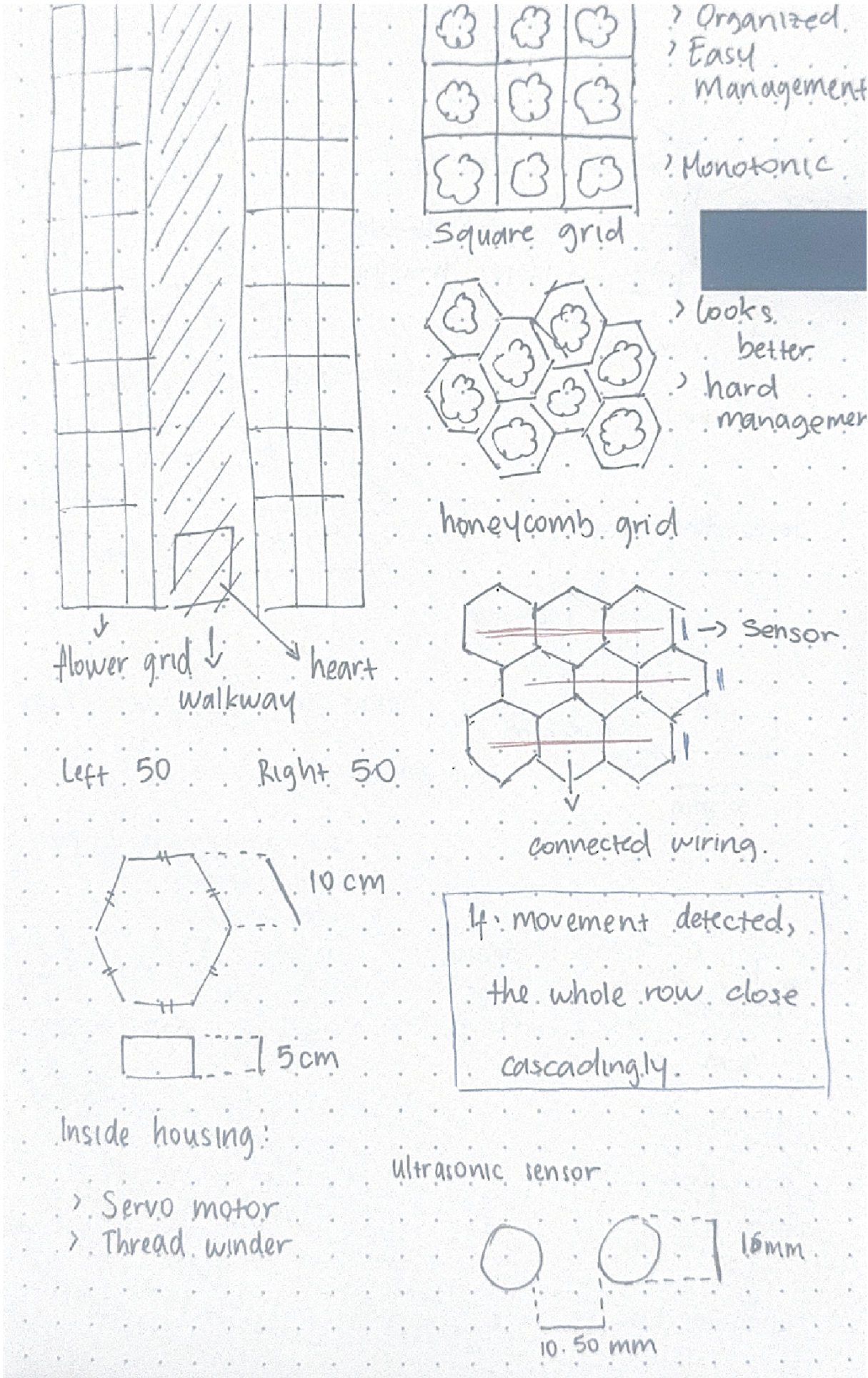
A Glimpse Into *My Thinking Process*

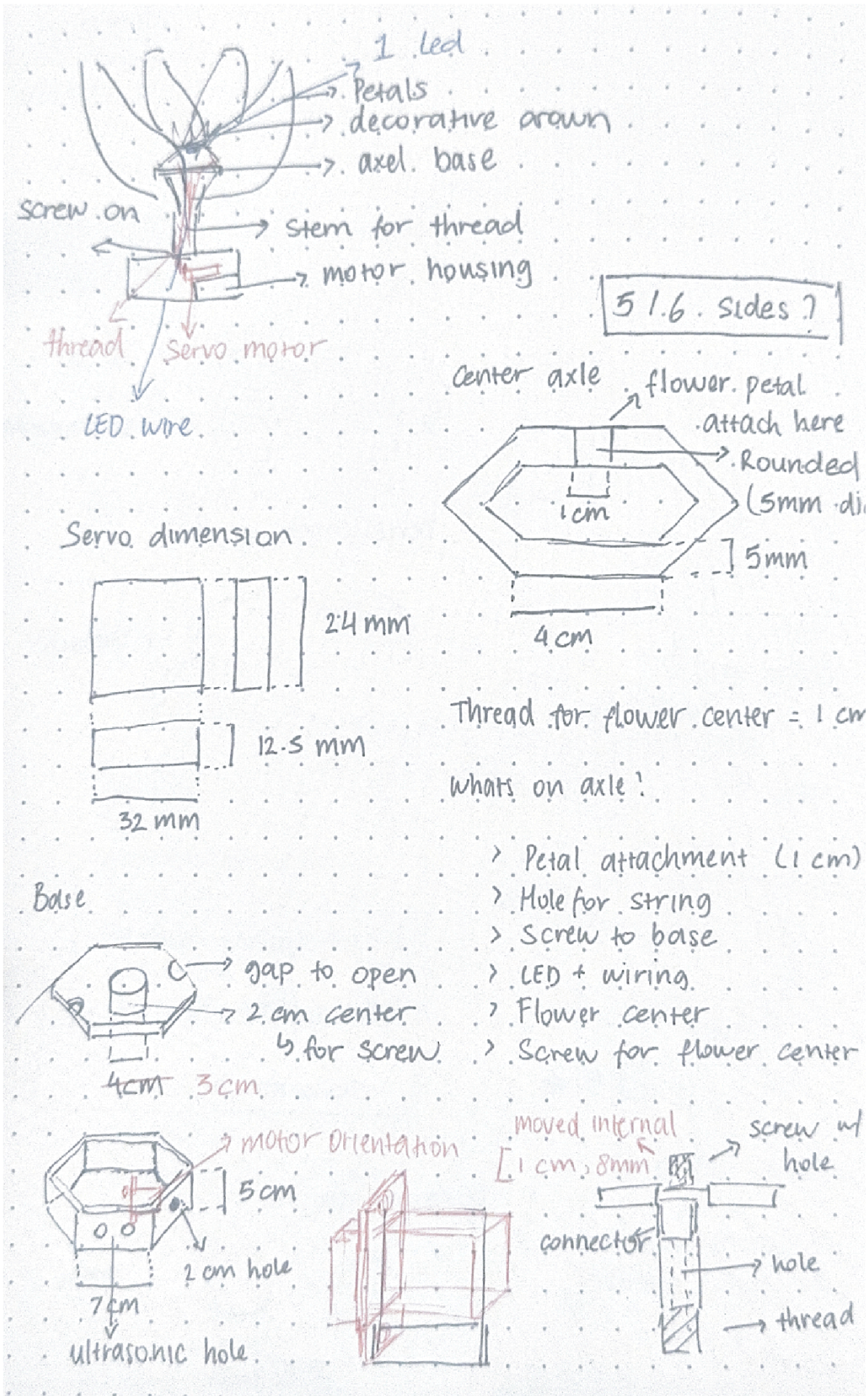
Carla Budar

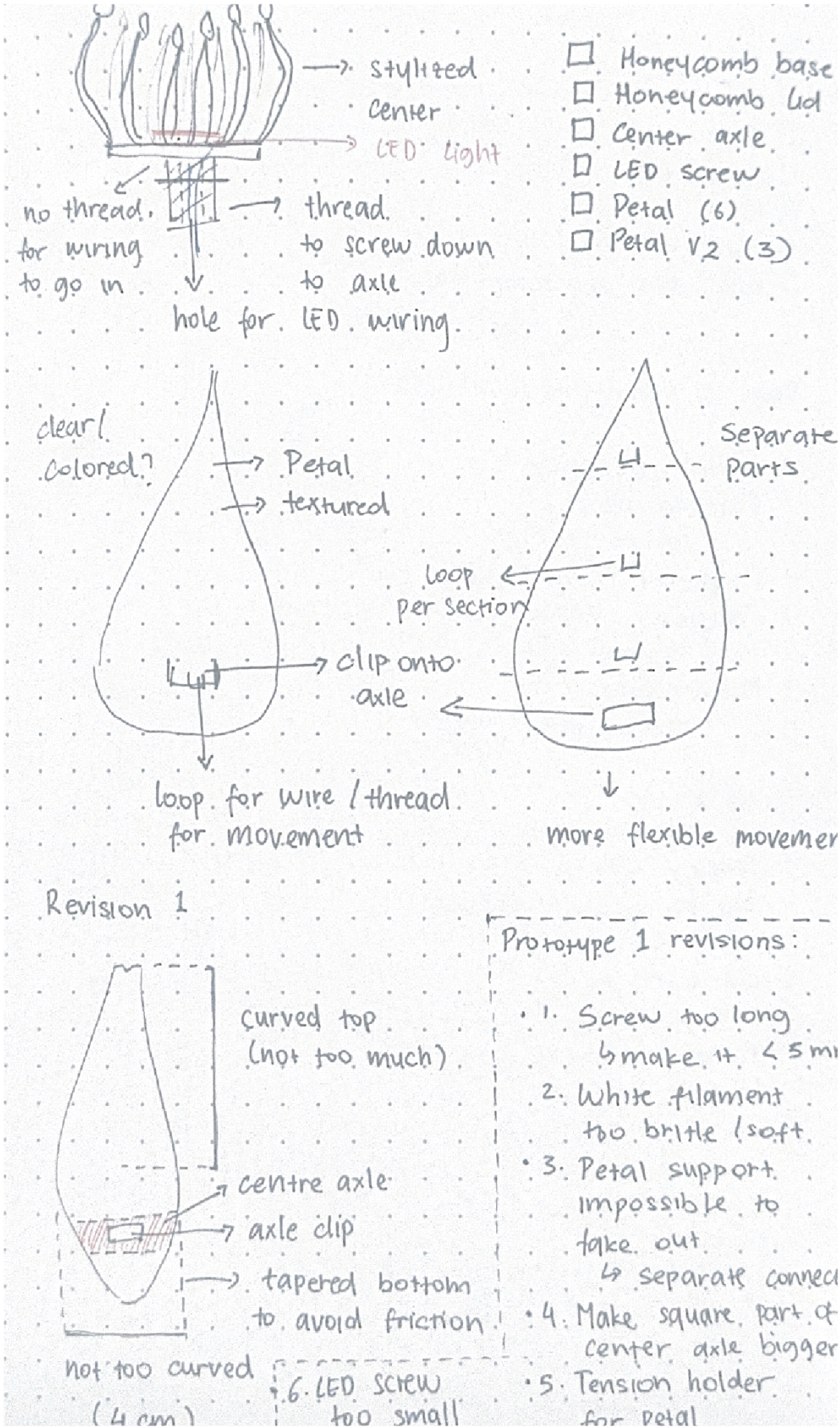
Parsons School of Design

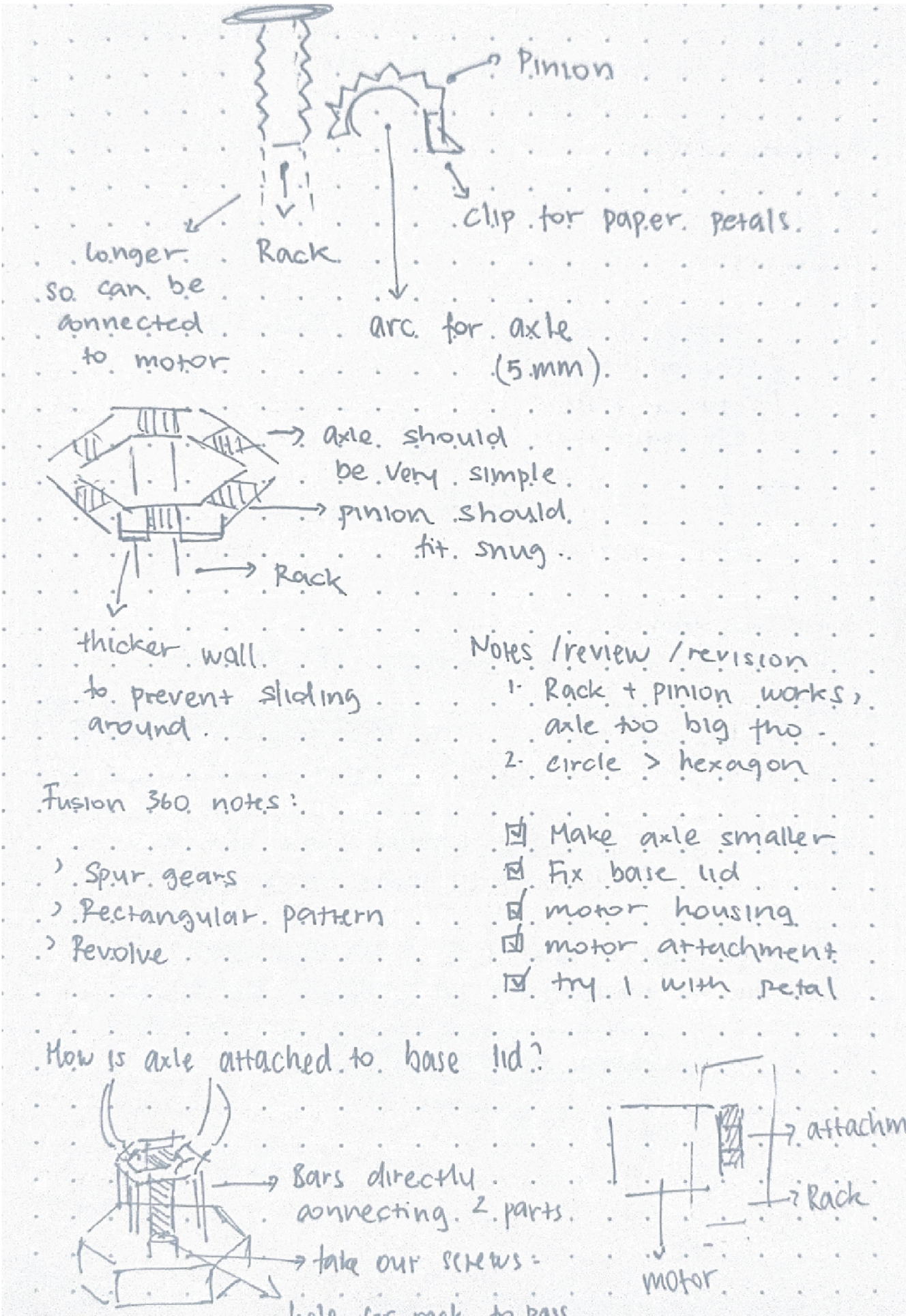
Spring 2026

Design and Technology









Print list :

- ☒ Lid
- ☒ New base ?
- ☒ motor housing
- ☒ pinion petal
- ☒ Motor attachment
- ☒ Center axle

→ Revision ?

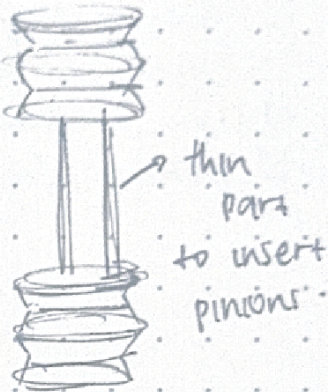
- axle too big (friction)
- housing needs cable hole
- Petal attachment reinforce
- Motor attachment hole too small

Plans for user testing.?

- > Test new mechanism.
- > User input

YAY Prototype 6.2 works.

- ↳ need to test gear torque



Prototype 7.

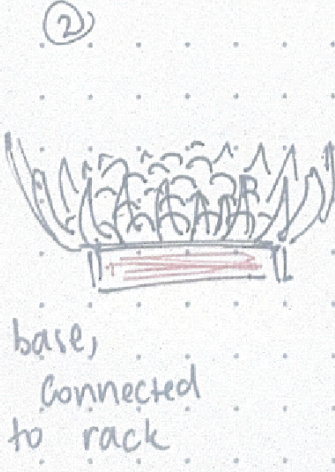
- > angled petal ✓
- > double layer ✓
- > center bud
- > gear attachment ✓
hole smaller
- > Rod through rack ✓
- > Motor housing placement ✓



①



②



Motor issues:

1. Not strong enough motor!
 - ↳ Petals drop when opening

2. Motor housing wall too thin

3. Pinion shifts

↳ moving rack away
from motor.

How to have more resistance ?

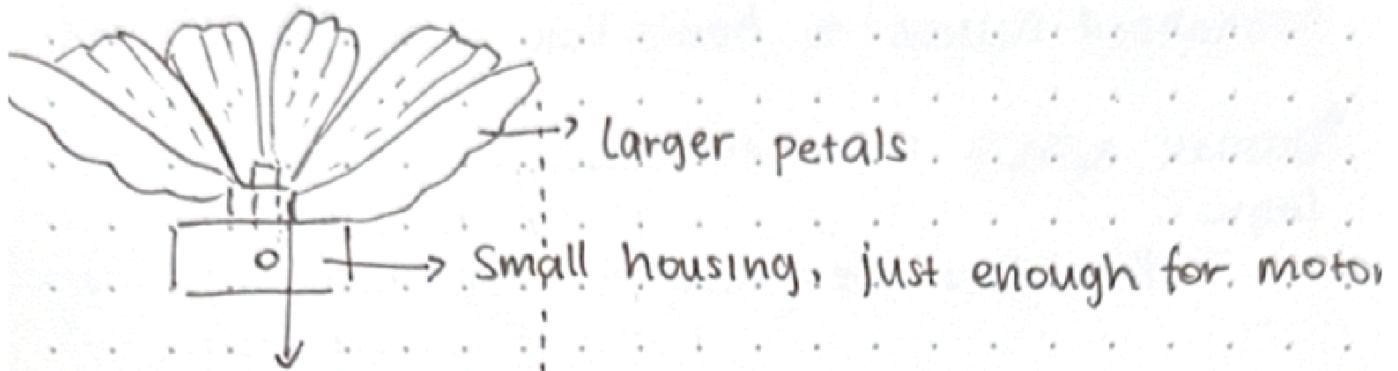
what to do?

1. Make motor housing walls thicker
2. Add 2x gear around rack
3. Make center post taller
4. Move housing for motor

→ Make movement more consistent

Prototype 9 revisions.

1. Make smaller gear. → If gears are closer together, petals would not drop open
2. Make bigger gear
3. Make motor attachment bigger so it moves / spins more



bigger rack to support larger + heavier petals
no more than 40 in

Petals + Pinion print together / separate?

Revisions:

- ☒ Rack (-) 1-2 mm (1/2) 0.7 mm
- ☐ add motor rack
- ☐ Motor attachments (M996) → should be big
- ☐ Housing
- ☐ Petal holder → thinner
- ☐ Small housing → remove us sensor hole

